

Erdős Problem #169

EP-169

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 3$ and $f(k)$ be the supremum of $\sum_{n \in A} \frac{1}{n}$ as A ranges over all sets of positive integers which do not contain a k -term arithmetic progression. Estimate $f(k)$.

Is $\lim_{k \rightarrow \infty} \frac{f(k)}{\log W(k)} = \infty$ where $W(k)$ is the van der Waerden number?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>